FAILURE OF ASSOCIATED POINTS TO LIE IN AN OPEN SUBSCHEME PERSISTS OVER A DENSE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6907-6921 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6923-6938, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Associated points of a module relative to a base, closures, and finite type spreading; the argument is retained.

Source reference. Original lemma 05AH: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6938–6938. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 054W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6496–6501. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05JV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6938–6938. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07T9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6938–6938. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $U \subset X$ be an open subscheme. Assume*

(1) *f is of finite type,*
(2) *$\mathcal{F}$ is of finite type,*
(3) *Y is irreducible with generic point η, and*
(4) *$Ass_{X_\eta}(\mathcal{F}_\eta)$ is not contained in U_η.*

Then there exists a nonempty open subscheme $V \subset Y$ such that for all $y \in V$ the set $Ass_{X_y}(\mathcal{F}_y)$ is not contained in U_y.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let $Z \subset X$ be the scheme theoretic support of $\mathcal{F}$, see Morphisms, Definition 05JV. Then Z_η is the scheme theoretic support of $\mathcal{F}_\eta$ (Morphisms, Lemma 07T9). Hence the generic points of irreducible components of Z_η are contained in $\mathrm{Ass}_{X_\eta}(\mathcal{F}_\eta)$ by Divisors, Lemma 05AH. Hence we see that $Z_\eta \cap U_\eta = \emptyset$. Thus $T = Z \setminus U$ is a closed subset of Z with $T_\eta = \emptyset$. If we endow T with the induced reduced scheme structure then $T \to Y$ is a morphism of finite type. By Lemma 054W there is a nonempty open $V \subset Y$ with $T_V = \emptyset$. Then V works. $\square$